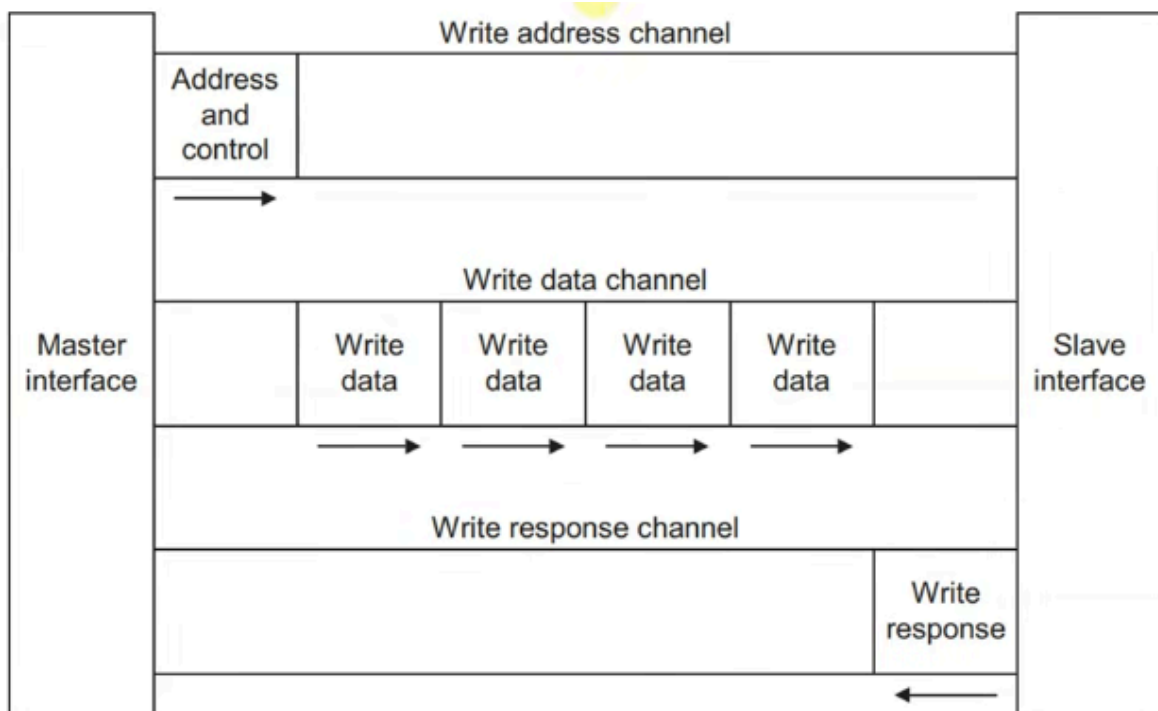
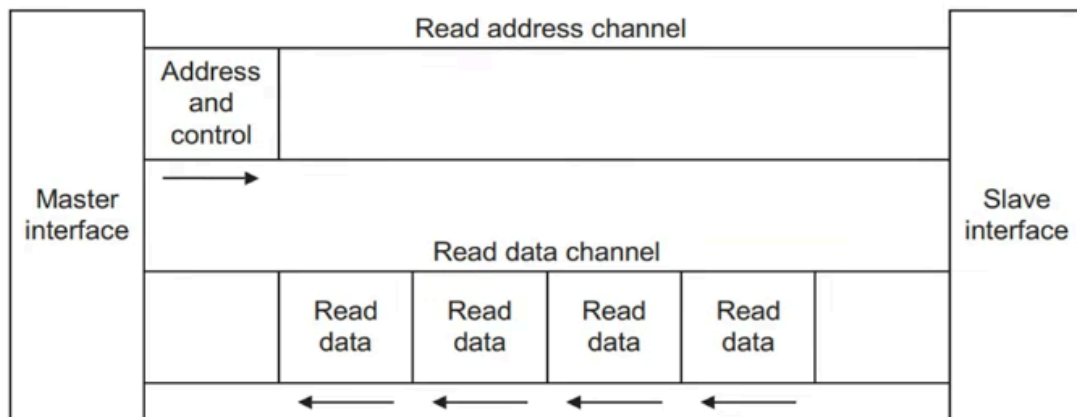
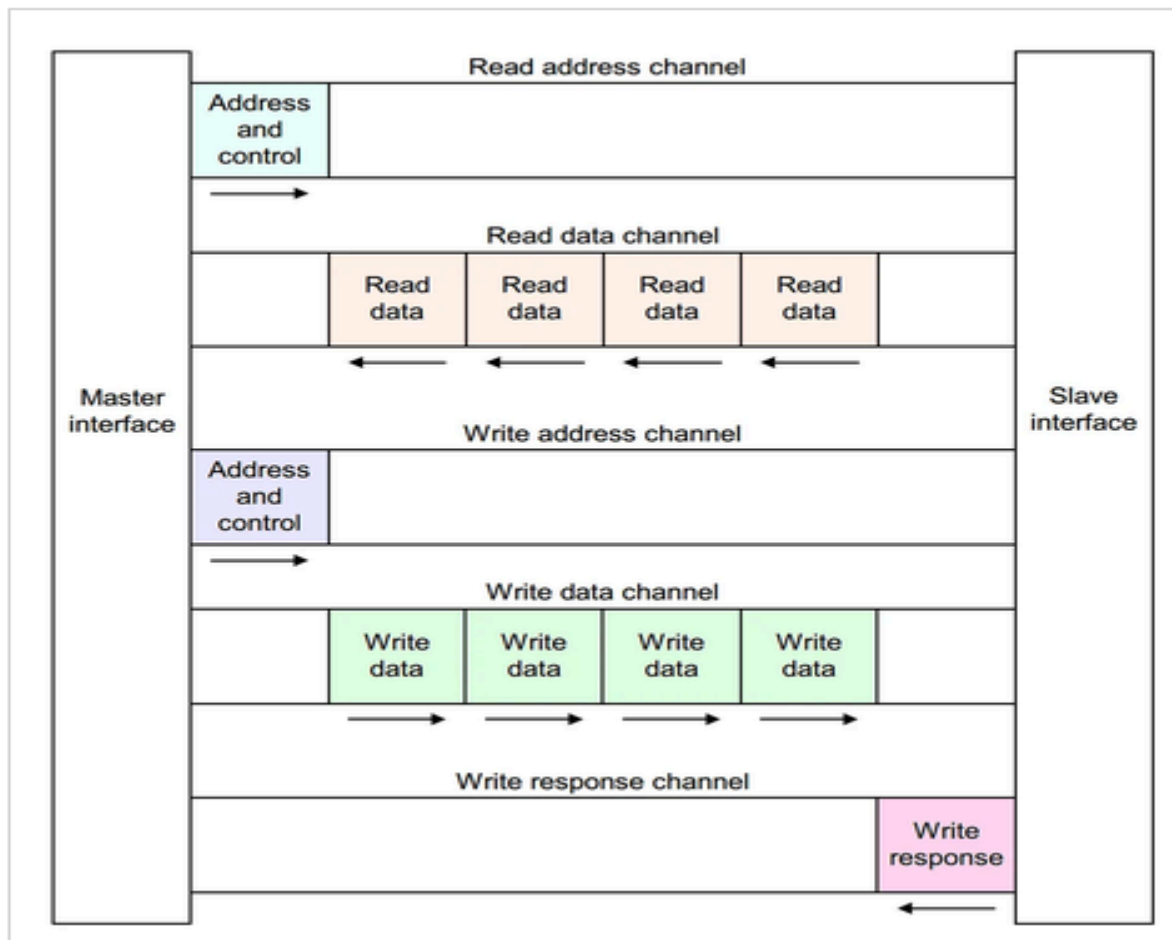






Master

Signal Name	Type	Width	Description	Channel	Direction	Handshake Signals
ACLK	input	1	Global Clock	Global	Input	N/A
ARESETN	input	1	Active Low Reset	Global	Input	N/A
START_READ	input	1	Start Read Transaction	Control	Input	N/A
START_WRITE	input	1	Start Write Transaction	Control	Input	N/A
address	input	ADDRESS	Target Address	Control	Input	N/A

W_data	input	DATA_WIDTH	Write Data	Control	Input	N/A
ARREADY	input	1	Read Address Channel Ready	Read Address	Input	ARREADY
RDATA	input	DATA_WIDTH	Read Data	Read Data	Input	N/A
RRESP	input	2	Read Response	Read Data	Input	N/A
RVALID	input	1	Read Data Valid	Read Data	Input	RVALID
AWREADY	input	1	Write Address Channel Ready	Write Address	Input	AWREADY
WREADY	input	1	Write Data Channel Ready	Write Data	Input	WREADY
BRESP	input	2	Write Response	Write Response	Input	N/A
BVALID	input	1	Write Response Valid	Write Response	Input	BVALID
ARADDR	output	ADDRESS	Read Address	Read Address	Output	N/A
ARVALID	output	1	Read Address Valid	Read Address	Output	ARVALID
RREADY	output	1	Read Data Ready	Read Data	Output	RREADY
AWADDR	output	ADDRESS	Write Address	Write Address	Output	N/A
AWVALID	output	1	Write Address Valid	Write Address	Output	AWVALID

WDATA	output	DATA_WIDTH	Write Data	Write Data	Output	N/A
WSTRB	output	4	Write Strobes	Write Data	Output	N/A
WVALID	output	1	Write Data Valid	Write Data	Output	WVALID
BREADY	output	1	Write Response Ready	Write Response	Output	BREADY

Slave

Signal Name	Type	Width	Description	Channel	Direction	Handshake Signals
ACLK	input	1	Global Clock	Global	Input	N/A
ARESETN	input	1	Active Low Reset	Global	Input	N/A
ARADDR	input	ADDRESS	Read Address	Read Address	Input	N/A
ARVALID	input	1	Read Address Valid	Read Address	Input	ARVALID
RREADY	input	1	Read Data Ready	Read Data	Input	RREADY
AWADDR	input	ADDRESS	Write Address	Write Address	Input	N/A
AWVALID	input	1	Write Address Valid	Write Address	Input	AWVALID
WDATA	input	DATA_WIDTH	Write Data	Write Data	Input	N/A
WSTRB	input	4	Write Strobes	Write Data	Input	N/A
WVALID	input	1	Write Data Valid	Write Data	Input	WVALID

BREADY	input	1	Write Response Ready	Write Response	Input	BREADY
ARREADY	output	1	Read Address Ready	Read Address	Output	ARREADY
RDATA	output	DATA_WIDTH	Read Data	Read Data	Output	N/A
RRESP	output	2	Read Response	Read Data	Output	N/A
RVALID	output	1	Read Data Valid	Read Data	Output	RVALID
AWREADY	output	1	Write Address Ready	Write Address	Output	AWREADY
WREADY	output	1	Write Data Ready	Write Data	Output	WREADY
BRESP	output	2	Write Response	Write Response	Output	N/A
BVALID	output	1	Write Response Valid	Write Response	Output	BVALID

Master

Current State	Transition Condition	Next State	Description
IDLE	START_READ	RADDR_CHANNEL	Start read operation
IDLE	START_WRITE	WRITE_CHANNEL	Start write operation
RADDR_CHANNEL	ARVALID && ARREADY	RDATA_CHANNEL	Read address handshake complete, move to read data phase

RDATA_CHANNEL	RVALID && RREADY	IDLE	Read data handshake complete, return to idle
WRITE_CHANNEL	AWVALID && AWREADY && WVALID && WREADY	WRESP_CHANNEL	Write address and data handshakes complete, move to response phase
WRESP_CHANNEL	BVALID && BREADY	IDLE	Write response handshake complete, return to idle
default	-	IDLE	Default state for any unspecified conditions

Slave

Current State	Transition Condition	Next State	Description
IDLE	START_READ	RADDR_CHANNEL	Start read operation
IDLE	START_WRITE	WRITE_CHANNEL	Start write operation
RADDR_CHANNEL	ARVALID && ARREADY	RDATA_CHANNEL	Read address handshake complete, move to read data phase
RDATA_CHANNEL	RVALID && RREADY	IDLE	Read data handshake complete, return to idle

WRITE_CHANNEL	AWVALID && AWREADY && WVALID && WREADY	WRESP_CHANNEL	Write address and data handshakes complete, move to response phase
WRESP_CHANNEL	BVALID && BREADY	IDLE	Write response handshake complete, return to idle
default	-	IDLE	Default state for any unspecified conditions